Chapter 1

Introduction to encryption, and perfect secrecy

> One of the most singular characteristics of the art of deciphering is the strong conviction possessed by every person, even moderately acquainted with it, that he is able to construct a cipher which nobody else can decipher.
>
> Charles Babbage, 1864

1.1 Secret communication and encryption schemes

Alice wants to send a message to Bob but not to Eve. But the channels Alice has at her disposal—broadcasting with radio waves, shouting across the room, or posting on a bulletin board—are easy for Eve to eavesdrop on. Alice does not want Eve to understand the message. Can we devise a method for Alice to send her message to Bob without Eve understanding it? Alice is often called the **sender** and Bob the **receiver**. Together, they are the **honest parties**. Eve is the **adversary** or **attacker**.

This is the problem of message **secrecy**. Alice can't simply send her message in the clear. Otherwise, Eve can simply read the message from the channel. Alice has to do something to the message before sending it. We call that process **encrypting**. Before it is encrypted, the **message** m is also called the **plaintext**. The result after encryption is called the **ciphertext** c. Alice sends the ciphertext over the insecure channel to Bob (and Eve). Once Bob receives the ciphertext, he will have to **decrypt** it to recover the plaintext.

The key

The first observation is that *Bob needs a secret that Eve doesn't know.* If Eve knows everything Bob knows, then Eve can decrypt the ciphertext herself exactly the same way Bob would—secret communication is impossible. We call this secret the **key** or **secret key** k.

Today we will focus on **symmetric encryption** (also called **secret-key encryption**). In this setting, Alice also knows the key k and the roles of sender and receiver are symmetric.[1] This is in contrast to **asymmetric encryption** (also called **public-key encryption**) which we will get to in a few lectures. Symmetric cryptography goes back a couple thousand years; asymmetric cryptography goes back only about 50 years.

Kerckhoffs' Principle

Auguste Kerckhoffs wrote the treatise "Military Cryptography, or Ciphers Used in Time of War, With a New Process of Decipherment Applicable to Double-Key Systems" in 1883.[2] In it, he laid out six requirements "of a cryptographic method destined to govern for an unlimited time the correspondence between different, army chiefs." One of these is so important it is now known simply as **Kerckhoffs' Principle**:

> It is necessary that [the cryptographic method] not demand secrecy, and that it may without inconvenience fall into enemy hands.

Here is how he explains it.

> In regard to the necessity for secrecy [of the method], which, in my opinion, constitutes the principal defect of all our systems of cryptography, I wish to point out that it restricts in some way the use of enciphered correspondence to only commanders in chief. And here I understand by secrecy, not the key properly so-called, but that which constitutes the material part of the systems: tableaux, code books, or whatever mechanical apparatus may be necessary for its application. In fact, it is not necessary to raise imaginary phantoms and to cast suspicion on the incorruptibility of employees or subordinate agents, in order to understand that, if a system requiring secrecy were found in the hands of too large a number of individuals, it could be jeopardized in any engagement in which one or another of them took part.
>
> ...
>
> [The Ministry for War] must absolutely renounce secret methods and establish as a policy that, it will only accept a process that can be taught in broad daylight in our military schools, that our students will be free to communicate to anyone they please, and that our neighbors can even copy and adopt if it seems suitable to them.

1.1.1 How to define a cryptographic scheme

A **cryptographic scheme** is a collection of algorithms meant to solve some problem. Typically, there are many ways to solve the same problem. For example, these notes describe a

[1] For most of human history, it was taken for granted that Alice and Bob must both know the key. Here's an (incorrect) argument. For any ciphertext c that Alice might send over the channel to Bob, there must be at least two possible plaintexts m_1 and m_2 that could result from decrypting with different keys. If not, then Eve would be able to decrypt c by just trying all keys. Alice must know which key Bob has in order to be sure that Bob can correctly recover the plaintext.

[2] Here is the translation I'm referencing: `https://archive.org/details/kerckhoffs.translated.pdf/page/5/mode/2up`.

few different schemes: shift cipher, substitution cipher, Vigenère cipher, and one-time pad. While these are different schemes, they all satisfy the same abstract template of a **symmetric encryption scheme**. To be able to say that one way of encrypting is better or worse than another, we must first specify what we want out of symmetric encryption schemes in the abstract.

There are typically three things that one must specify when defining a cryptographic scheme: syntax, correctness, and security. The **syntax** specifies what algorithms are required, along with their inputs and outputs. This sometimes involves dealing with a few technicalities, but is generally not too complex. **Correctness** specifies what functionality the algorithms must satisfy when used by honest parties. This is typically straightforward. **Security** specifies what guarantees are required in the face of an adversary. Defining security is subtle, and a topic we will spend a lot of time on in this class.

1.1.2 Symmetric encryption schemes (information theoretic)

Let's apply this to symmetric encryption, by defining syntax and correctness. We'll leave security for later. An encryption scheme is defined with respect to three non-empty sets: the **message space** $\mathcal{M}$, the **ciphertext space** $\mathcal{C}$, and the **key space** $\mathcal{K}$. These are the sets of possible plaintexts, ciphertexts, and secret keys respectively. We require the key space $\mathcal{K}$ to be *finite* so that uniform sampling is well-defined.

Remark 1.1. The syntax will need slight modification when we move from *information theoretic* security (without computational hardness assumptions) to *computational* security (with computational hardness assumptions).

Definition 1.1 (Symmetric encryption scheme: syntax (information theoretic))**.** *A* ***symmetric encryption scheme*** *is a pair of algorithms* $(\mathsf{Enc}, \mathsf{Dec})$ *defined with respect to sets* $\mathcal{M}$*,* $\mathcal{C}$*, and* $\mathcal{K}$ *such that:*

$\mathsf{Enc} : \mathcal{K} \times \mathcal{M} \to \mathcal{C}$**.** *The* encryption algorithm Enc *takes as input a key* k *and a message* m*. It outputs a ciphertext* $c \leftarrow \mathsf{Enc}_k(m)$*.* Enc *may be randomized.*

$\mathsf{Dec} : \mathcal{K} \times \mathcal{C} \to \mathcal{M}$**.** *The* decryption algorithm Dec *takes as input a key* k *and a ciphertext* c*. It outputs a message* $m = \mathsf{Dec}_k(c)$*.* Dec *is deterministic.*[3]

We always assume that keys are ***sampled uniformly at random*** *from the key space:* $k \xleftarrow{\$} \mathcal{K}$*.*

Definition 1.2 (Symmetric encryption: correctness)**.** *A symmetric encryption scheme* $(\mathsf{Enc}, \mathsf{Dec})$ *is* ***correct*** *if for all* $k \in \mathcal{K}$*,* $m \in \mathcal{M}$*:*

$$\Pr\left[\mathsf{Dec}_k(\mathsf{Enc}_k(m)) = m\right] = 1.$$

When we talk about a concrete encryption scheme, it will sometimes be useful to give the scheme as a whole a name. A common generic name for a cryptographic scheme is Π. If Π is an encryption scheme, then Π is just shorthand for a pair of algorithms $\Pi = (\mathsf{Enc}, \mathsf{Dec})$. If we need

[3] Correctness implies that requiring Dec to be deterministic is without loss of generality (see Exercise 1.4).

to name these particular algorithms—for example, to distinguish the algorithms of Π from another scheme Π'—we can write $\Pi.\mathsf{Enc}$ and $\Pi.\mathsf{Dec}$. Likewise for the message/ciphertext/key spaces ($\Pi.\mathcal{M}$, $\Pi.\mathcal{C}$, $\Pi.\mathcal{K}$).

Example 1.1 (Correct-but-insecure encryption)**.** For any message space $\mathcal{M}$, let $\mathcal{C} = \mathcal{M}$ and consider the encryption scheme with $\mathcal{K} = \{\diamondsuit, \clubsuit, \heartsuit, \spadesuit\}$ and $(\mathsf{Enc}, \mathsf{Dec})$ defined as follows:

$\mathsf{Enc}_k(m)$
return m

$\mathsf{Dec}_k(c)$
return c

This encryption scheme is obviously insecure: the ciphertext is exactly the message. This encryption scheme is also correct: for all $k \in \mathcal{K}$, $m \in \mathcal{M}$:

$$\mathsf{Dec}_k(\mathsf{Enc}_k(m)) = \mathsf{Dec}_k(m) = m.$$

1.2 Some historical encryption schemes (pre-1900)

Let's see some example encryption schemes. Let $\mathsf{Alpha} = \{A, B, C, \ldots, Z\}$ be the set of all uppercase characters in the English alphabet, and let Alpha^+ be the set of all finite nonempty strings of English characters.

Shift cipher

For a character $x \in \mathsf{Alpha}$ and number $n \in \mathbb{Z}$, define the function $\mathsf{shift}(x, n)$ that shifts x by n places in the alphabet (i.e., $A \to B \to C \to \cdots \to Z \to A$) and returns the result. For example, $\mathsf{shift}(A, 1) = B$, $\mathsf{shift}(Z, 4) = D$, and $\mathsf{shift}(A, -2) = Y$. Observe that $\mathsf{shift}(x, n) = \mathsf{shift}(x, n \bmod 26)$. So $\mathsf{shift}(A, 26) = A$ and $\mathsf{shift}(A, 27) = B$.

The shift cipher $\mathsf{SHIFT} = (\mathsf{Enc}, \mathsf{Dec})$ has message and ciphertext spaces $\mathcal{M} = \mathcal{C} = \mathsf{Alpha}^+$ and key space $\mathcal{K} = \{0, 1, \ldots, 25\}$. Encryption and decryption are as follows.

$\mathsf{SHIFT.Enc}_k(m_1 \ldots m_\ell)$
for $i = 1, \ldots, \ell$ **do** $c_i \leftarrow \mathsf{shift}(m_i, k)$ **return** $c_1 c_2 \ldots c_\ell$

$\mathsf{SHIFT.Dec}_k(c_1 \ldots c_\ell)$
for $i = 1, \ldots, \ell$ **do** $m_i \leftarrow \mathsf{shift}(c_i, -k)$ **return** $m_1 m_2 \ldots m_\ell$

Example 1.2. $\mathsf{SHIFT.Enc}_7(\texttt{HELLOWORLD}) = \texttt{OLSSVDVYSK}$.

It is obvious that SHIFT is correct (Def. 1.2). (Then again, in cryptography "obvious truths" often turn out false.) Let's give a proof to warm ourselves up.[4] Because SHIFT.Enc and SHIFT.Dec are both deterministic, we can omit any probability notation.

[4] You may be a bit rusty with proofs. If so, take this as a chance to get your bearings—things are only going to get harder (e.g., Theorem 1.1 below). Notice the structure of the proof: it starts by stating what must be

Claim 1.1. SHIFT *is a correct symmetric encryption scheme.*

Proof. Fix arbitrary key k and message m. Let $m' = \mathsf{Dec}_k(\mathsf{Enc}_k(m))$. We must show that for all $i \in \{1, \ldots, |m|\}$: $m'_i = m_i$. Expanding the definitions of Enc and Dec, we observe that $m'_i = \mathsf{shift}(\mathsf{shift}(m_i, k), -k) = \mathsf{shift}(m_i, 0) = m_i$. □

Substitution cipher

The substitution cipher SUB has message space and ciphertext space $\mathcal{M} = \mathcal{C} = \mathsf{Alpha}^+$. The key space is the set of all permutations over Alpha. That is, a key is a permutation (one-to-one and onto function) $\phi : \mathsf{Alpha} \to \mathsf{Alpha}$. $\mathcal{K}$ is the set of all such permutations. Encryption and decryption are as follows.

$\mathsf{SUB.Enc}_\phi(m_1 \ldots m_\ell)$

for $i = 1, \ldots, \ell$ **do**
  $c_i \leftarrow \phi(m_i)$
return $c_1 c_2 \ldots c_\ell$

$\mathsf{SUB.Dec}_\phi(c_1 \ldots c_\ell)$

for $i = 1, \ldots, \ell$ **do**
  $m_i \leftarrow \phi^{-1}(c_i)$
return $m_1 m_2 \ldots m_\ell$

Example 1.3. Suppose a uniformly sampled permutation ϕ is given by the following table:

x	A	B	C	D	E	F	G	H	I	J	K	L	M
$\phi(x)$	O	P	W	Z	Y	I	G	T	S	Q	V	F	H

x	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
$\phi(x)$	C	X	L	R	M	B	U	N	E	A	D	J	K

Applying ϕ to each character gives

m_i	H	E	L	L	O	W	O	R	L	D
$\phi(m_i)$	T	Y	F	F	X	A	X	M	F	Z

Thus $\mathsf{SUB.Enc}_\phi(\texttt{HELLOWORLD}) = \texttt{TYFFXAXMFZ}$. Reading the permutation table in reverse gives ϕ^{-1}, so decrypting TYFFXAXMFZ recovers HELLOWORLD.

Claim 1.2. SUB *is a correct symmetric encryption scheme.*

Proof. Exercise 1.2. □

Vigenère cipher

The Vigenère cipher VIG has message space and ciphertext space $\mathcal{M} = \mathcal{C} = \mathsf{Alpha}^+$. The key space is $\{0, 1, \ldots, 25\}^\ell$ for some parameter $\ell \geq 1$ called the *key length*. Encryption and decryption are as follows.

shown ($m'_i = m_i$) and concludes by showing it. All variables are defined upon first use. The proof allows k and m to be arbitrary, as is required by the "for all" quantifiers in Def. 1.2. There are no worked examples, nor any other unnecessary commentary.

Write the key as $k = (k_1, k_2, \ldots, k_\ell)$. The key index cycles through $1, \ldots, \ell$, so k_1 is used at positions $1, \ell + 1, 2\ell + 1, \ldots$.

$\mathsf{VIG.Enc}_k(m_1 \ldots m_L)$

for $i = 1, \ldots, L$ **do**
  $j \leftarrow [(i-1) \bmod \ell] + 1$
  $c_i \leftarrow \mathsf{shift}(m_i, k_j)$
return $c_1 c_2 \ldots c_n$

$\mathsf{VIG.Dec}_k(c_1 \ldots c_n)$

for $i = 1, \ldots, L$ **do**
  $j \leftarrow [(i-1) \bmod \ell] + 1$
  $m_i \leftarrow \mathsf{shift}(c_i, -k_j)$
return $m_1 m_2 \ldots m_n$

Example 1.4. Let $\ell = 3$ and $k = (k_1, k_2, k_3) = (1, 2, 3)$, and encrypt `HELLOWORLD`. The shifts start with k_1 and repeat every three letters:

i	1	2	3	4	5	6	7	8	9	10
m_i	H	E	L	L	O	W	O	R	L	D
shift	1	2	3	1	2	3	1	2	3	1
c_i	I	G	O	M	Q	Z	P	T	O	E

Thus $\mathsf{VIG.Enc}_{(1,2,3)}(\texttt{HELLOWORLD}) = \texttt{IGOMQZPTOE}$. The shift $k_1 = 1$ is used at positions $1, 4, 7, 10$. Observe that repeated characters in the plaintext may be shifted into different characters in the ciphertext. For example, `L` becomes `O`, `M`, and `O`.

Claim 1.3. VIG *is a correct symmetric encryption scheme (for every key length ℓ).*

Proof. Exercise 1.2. □

1.2.1 Every pre-1900 encryption scheme is insecure

Obviously we just scratched the surface of historical encryption schemes. Let's see why the schemes above are insecure (without having yet defined security).

Brute force search (Shift cipher)

The shift cipher is easy to break in practice. Consider the ciphertext

$$c = \texttt{AOPZPZHYLHSSFKPMMPJBSATLZZHNLAVKLJYFWAQBSPBZJHLZHY.}$$

We can compute all possible plaintexts by **brute force search** over the possible keys: just try decrypting with each key. If there are few enough keys and the messages are structured, then it may be possible to identify the right one.[5]

[5] Looking ahead, brute force search will fail against the one-time pad because there is no way to check which candidate decrypted message is correct: every ciphertext can decrypt to every possible message.

Shift	Message
1	ZNOYOYGXKGRREJOLLOIARZSKYYGMKZUJKIXEVZPAROAYIGKYGX
2	YMNXNXFWJFQQDINKKNHZQYRJXXFLJYTIJHWDUYOZQNZXHFJXFW
3	XLMWMWEVIEPPCHMJJMGYPXQIWWEKIXSHIGVCTXNYPMYWGEIWEV
4	WKLVLVDUHDOOBGLIILFXOWPHVVDJHWRGHFUBSWMXOLXVFDHVDU
5	VJKUKUCTGCNNAFKHHKEWNVOGUUCIGVQFGETARVLWNKWUECGUCT
6	UIJTJTBSFBMMZEJGGJDVMUNFTTBHFUPEFDSZQUKVMJVTDBFTBS
7	THISISAREALLYDIFFICULTMESSAGETODECRYPTJULIUSCAESAR ←
8	SGHRHRZQDZKKXCHEEHBTKSLDRRZFDSNCDBQXOSITKHTRBZDRZQ
	...
26	AOPZPZHYLHSSFKPMMPJBSATLZZHNLAVKLJYFWAQBSPBZJHLZHY

There are two important lessons here. First, security (whatever that means) requires the key space to be large enough that brute force search is infeasible. Second, messages are not typically arbitrary or random strings. They have structure (e.g., meaningful English text) that an attacker can take advantage of.

★ Frequency analysis (Substitution & Vigenère ciphers)

The example above shows us that a very large key space is necessary for security. **Large key space is not sufficient for security**, though it is a common mistake to believe that it is.[6] See Exercise 1.1.

The idea, going back to Al-Kindi in the 9th century, is that in natural language, different characters appear with very different frequencies. Moreover, these frequencies are remarkably common across texts. In English, E is the most common letter, followed by T and A. The observation extends to combinations of letters (TH, ER). This is the basis of **frequency analysis**. This is where the second lesson above—an attacker can take advantage of a plaintext's structure—really shows its importance.

In a substitution cipher, every appearance of a given character in the plaintext is transformed into the same character in the ciphertext (and vice versa). Therefore, for a ciphertext encrypting natural English text of moderate length, the most common letter in the ciphertext is very likely to correspond to Es in the plaintext, and so on. Wikipedia has a good worked example of frequency analysis in action.[7]

At first glance, the Vigenère cipher appears immune to frequency analysis. For a key of length ℓ, a given character in the plaintext may be transformed into ℓ different characters in the ciphertext.[8] The observation needed to break the Vigenère cipher is very clever: every ℓth character of the ciphertext is the result of shifting plaintext characters by the same amount. Consider the characters at positions 1, $\ell+1$, $2\ell+1$, ..., in the plaintext. This subsequence of characters will have the single-character frequencies of English. Same with the characters at

[6] For example, in an 1801 letter to Thomas Jefferson, mathematician Robert Patterson describes an encryption scheme that he believes unbreakable: "On calculating the number of changes, and combinations, of which the above cypher is susceptible, ... it will be found to amount to upwards of ninety millions of millions, nearly equal to the number of seconds in three millions of years! Hence I presume the utter impossibility of decyphering will be readily acknowledged." `https://founders.archives.gov/documents/Jefferson/01-36-02-0093`.

[7] `https://en.wikipedia.org/wiki/Frequency_analysis`.

[8] It is an example of a *polyalphabetic* cipher, in contrast to *monoalphabetic* ciphers like the shift and substitution ciphers.

positions 2 mod ℓ, 3 mod ℓ, and so on. Frequency analysis can be applied to each subsequence separately.

A fun puzzle: How can you use frequency analysis to discover the key length ℓ? Try to decrypt the message in the footnote.[9]

A game of cat and mouse

For about 2000 years, cryptography was a cat-and-mouse game between those who developed new schemes and those who broke them. This culminated in the Enigma machine, used by the Germans to encrypt their most sensitive communications during World War II. It was famously broken, originally by Polish cryptanalysts and, building on their work, by British cryptanalysts including Alan Turing.

Some considered this state of affairs inevitable. For example, Kerckhoffs considered provable security impossible. In "Military Cryptography" (1883), he writes:

> ...it is hardly possible to establish a completely indecipherable system. It must be understood: I know very well that to try to find, under these conditions, a mathematically indecipherable system is something mathematically impossible, but I affirm, and not without good reasons, that while bringing about the realization of the different desiderata that I have enumerated above, it is perfectly possible to compose systems that are, if not mathematically, at least practically indecipherable.

1.3 Our first security definition: Perfect Secrecy

For 2000 years, encryption was a game of cat and mouse. Cryptographers would design encryption schemes they didn't know how to break, and cryptanalysts would break them. The failure of various attempts to prove schemes mathematically unbreakable led some to conclude that the cat-and-mouse game was inherent, that empiricism was the only basis for security analysis.

That changed in 1949, when Claude Shannon published "Communication Theory of Secrecy Systems." In it, he gave a mathematical **definition of security** called **perfect secrecy**, and proved that a scheme called the **one-time pad** satisfies that definition. He was the right guy for the job, having just invented information theory.

1.3.1 Priors and posteriors

Here is how Shannon described his idea.

[9] UGSWQCFJBNVAVYFJXOKDPFFEWAEPWINDIUZWSWQFBUMHIFLLGZYMKSWIBTZBTPGCLEJLYTMSHPJGTNLVIJJIYELVPEVQRGTPWNSBZIRMGTPFFSTNLLPWRKCDBSDRHPGFODWLM
OJHFADUMDFRYTPWESSZLETMSSPEVTILDMLSCYDXATXJHKMAVVBZDMYITRAUHIWTNLYWVRVQFADAUHIWMHILQLVVMSHNSMOEJHDLADACZXFADAUHIWYRWEBZDYUDDMTETBQWGWVKSKGY
PMGTXJHKMDIQIFCOBQWZGWOFVLHMUIDKZXZBFWFBOGMHMWDWMDHSTMSFIFPHRVAVYVZLQMGTRJSVMDIQIFCBIPLZXPMFWMHMJMTDVZUKLPYFOWBNBZMEHYMFHEITFXDMEZKIFCGYFBS
SUWJGXTPWESSZLEYQDYUIHPABWZKZIXSHCBMEFWAENAZEZHYZMZCPEJYBDMVBZDRUFXQHIIJFAWMJMMMYYDMGTRJSPTMMFBXQJGFADLMUFULWPAKZVZLQTADHFUKXFQJUSLZHFTMSUU
XDLSWSVVFJXOTKAQEUKXFQJUSLZHFADPHFODGDBZMWUZHUGFPZEUKXMWJVAMBQQKDITFTHVOVVLSXVHPZNSEBJGEEBLPWVVNQKRJZEFUMHMZMSUZHNXFPFIFUXDBGOWSCYDNMIAPOHI
LIUMSMYFQMSWQESBEAVVIHOZUDTMSUUXDLSWSVVFJXOTKAQEUKXDZQTSMYYMKSWMOEWAEOSBZDMCZZSDSFUKXRWXBZDRUFXQHOBMOXDPWAWZNUZWFDPTBZMHILQLVVMSHNSMOEJHDAS
QVKZNFADTMSUKURQFOXNMNTZQPETUKXHMJJQHZFPBMVEFFGTNLLPWEMOUMSGQFZLXLLAVYEMOUMZUFFSKBSSAVVVCIEXRTQEJVBNQLAWKAOBHMITFFDKTPSVVHOQMLRDMOEWAEMSZLG
WLANFWFGPUMHOJIKRVHPADGNZJHEDQFOKDZXMYSTDIJVDIVVIFCOBQMQTQZJHEDQFOXQPCFPGDEFTHXDESAAMDNEXKUMGUHKHQKSAMYUZWFDPTBZMHILQLVVMSHNSMOEWAEMNMFHIAM
GCITFNRKNQFOODMYFADITJSGWAGSVVFJXETHSXFUWAEZWJWKDATMRXZUIHYIZEIEDINAYSWQIFDOEVLWVHQCPXSWQEBBYRWEBZDICSASPZEMHMTPWUTDAIDLHVZTBQWFWJAWZNIZLZC
PGPUWAGKIFCTYMKRPZEMHMTPWUTDAIDEHVTUTLGTPWNAQHUYXMIAGUKXHMSDWMOISBUTXJHKMUXGVLGZYMKSWMOELMWIKAGZIXSHCBMEFWPOOJMSSGCSASHFIFJKEILMJKDATMSDDVM
HMHMVIQZIXFADAQTTHKLQYPLSJLGEDITFOLZHBZMEZYYFADHFBSVTLAGIFCBIPLDIFIFPBNBZMXHMGMFDCFPGWAEPWINDINAZHKQMJJATCHWFSCYQTQITBOGMOZMTWNQYDMGTPBZDGD
WNMJSCYZBFWFBOGMOLADACZNTXKXSIUIKOULPWCVLWGDHEBOGZOLKIOSCUFBSLMTHRHDIFLLGZYHXMXZHBQWTPWUGQICZZVTDFUKXFWMZLGYUKTMSSPEVTILDMLSCYITSTDTCUBNOXW
JSCUNNMSMOUORTPWUGUDHSVQTMUVUXTPSBZZOBXBETMOEIHWTLPSSHUKYKNMCPYXTPWMSQOBUGSWQPQHGFQJUSLZHFHEWQBWHGAVVOGCXLQT

> "Perfect Secrecy" is defined by requiring of a system that after a cryptogram is intercepted by the enemy the a posteriori probabilities of this cryptogram representing various messages be identically the same as the a priori probabilities of the same messages before the interception.

"In this case," he concludes, "intercepting the message has given the cryptanalyst no information." Let's make these ideas formal.

Consider a random variable M over $\mathcal{M}$ distributed according to the *a priori* probability of each message m: $\Pr[M = m]$. One useful interpretation is that M reflects the beliefs that the attacker has about the unknown message m before he sees the ciphertext c. For example (spaces for readability):

$$M = \begin{cases} \texttt{SEL NVDA} & \text{with probability } 1/3 \\ \texttt{BUY NVDA} & \text{with probability } 1/2 \\ \texttt{BUY AAPL} & \text{with probability } 1/6. \end{cases}$$

Now suppose we encrypt M using a symmetric encryption scheme: we sample $k \xleftarrow{\$} \mathcal{K}$ uniformly at random and compute $c \leftarrow \mathsf{Enc}_k(M)$. After observing the realized ciphertext c, the attacker can compute the *a posteriori* probability of each message $\Pr[M = m \mid \mathsf{Enc}_k(M) = c]$.

Example 1.5. Consider $\Pi = \mathsf{SUB}$ and messages distributed as above. Suppose you observe the ciphertext $c = \texttt{WNX TTAD}$. What can you conclude about the plaintext?

There is a repeated character in the fourth and fifth positions of c. For the substitution cipher, repeated characters in the ciphertext always correspond to repeated characters in the plaintext. Thus

$$\Pr_{M,k \xleftarrow{\$} \mathcal{K},\mathsf{Enc}} [M = \texttt{BUY AAPL} \mid \mathsf{Enc}_k(M) = \texttt{WNX TTAD}] = 1.$$

By similar reasoning (see Exercise 1.6),

$$\Pr[M = \texttt{BUY AAPL} \mid \mathsf{Enc}_k(M) = \texttt{NCO QPEZ}] = 0.$$

Moreover, some ciphertexts are impossible:

$$\Pr[\mathsf{Enc}_k(M) = \texttt{AAA BBBB}] = 0.$$

1.3.2 Definition of perfect secrecy

Perfect secrecy requires that the prior and posterior distributions are always identical, regardless of what (possible) ciphertext is observed and regardless of the distribution over messages. It implies that M and $\mathsf{Enc}_k(M)$ are independent random variables.

Recall that the key is assumed to be uniform over the key space: $k \xleftarrow{\$} \mathcal{K}$.

Definition 1.3 (Perfect secrecy)**.** *A symmetric encryption scheme is* ***perfectly secret*** *if for all random variables M over $\mathcal{M}$, all messages $m \in \mathcal{M}$, and all ciphertexts $c \in \mathcal{C}$:*[10]

$$\Pr_{M,k \xleftarrow{\$} \mathcal{K},\mathsf{Enc}}[M = m \mid \mathsf{Enc}_k(M) = c] = \Pr_{M}[M = m] \quad \textit{or} \quad \Pr_{M,k \xleftarrow{\$} \mathcal{K},\mathsf{Enc}}[\mathsf{Enc}_k(M) = c] = 0.$$

Perfect secrecy implies that the ciphertext reveals no *partial information* about the message. For instance, the attacker in the previous example doesn't learn whether stocks are being bought or sold, nor whether the message pertains to NVDA or AAPL stock.

1.3.3 An equivalent condition: perfect indistinguishability

Unfortunately, perfect secrecy is not easy to actually work with. The "for all M" quantifier is cumbersome (as you'll see in the proof of Theorem 1.1). Fortunately, Shannon gave an equivalent condition that is much easier to work with: that every pair of messages is mapped to a given ciphertext with equal probability. We will call this condition **perfect indistinguishability** (there is no standard name).

Definition 1.4 (Perfect indistinguishability)**.** *A symmetric encryption scheme satisfies* ***perfect indistinguishability*** *if for all $m_1, m_2 \in \mathcal{M}$ and all $c \in \mathcal{C}$:*

$$\Pr_{k \xleftarrow{\$} \mathcal{K},\mathsf{Enc}}[\mathsf{Enc}_k(m_1) = c] = \Pr_{k \xleftarrow{\$} \mathcal{K},\mathsf{Enc}}[\mathsf{Enc}_k(m_2) = c].$$

Theorem 1.1
A symmetric encryption scheme satisfies perfect secrecy if and only if it satisfies perfect indistinguishability.

Proof. Let $\Pi = (\mathsf{Enc}, \mathsf{Dec})$ be a symmetric encryption scheme.

(PS $\Rightarrow$ PI) Suppose Π is perfectly secret. Let $m_1, m_2 \in \mathcal{M}$ and $c \in \mathcal{C}$. Let M be a random variable over $\mathcal{M}$ such that $\Pr[M = m_1] > 0$ and $\Pr[M = m_2] > 0$. We must show that

$$\Pr[\mathsf{Enc}_k(m_1) = c] = \Pr[\mathsf{Enc}_k(m_2) = c]. \tag{1.1}$$

We consider two cases, depending on whether $\Pr[\mathsf{Enc}_k(M) = c] = 0$.

Case 1: Suppose $\Pr[\mathsf{Enc}_k(M) = c] = 0$. By the Law of Total Probability

$$\begin{aligned} 0 = \Pr[\mathsf{Enc}_k(M) = c] &= \sum_{m:\Pr[M=m]>0} \Pr[M = m] \cdot \Pr[\mathsf{Enc}_k(M) = c \mid M = m] \\ &= \sum_{m:\Pr[M=m]>0} \Pr[M = m] \cdot \Pr[\mathsf{Enc}_k(m) = c]. \end{aligned}$$

[10] Recall that $\Pr[A|B] = \frac{\Pr[A \text{ and } B]}{\Pr[B]}$ is undefined when $\Pr[B] = 0$.

The last equality is by simplifying $\Pr[\mathsf{Enc}_k(M) = c \mid M = m] = \Pr[\mathsf{Enc}_k(m) = c \mid M = m] = \Pr[\mathsf{Enc}_k(m) = c]$, using the fact that M is independent of the randomness used to sample k and to run Enc.

Each term in the sum must equal 0. Because $\Pr[M = m_1] > 0$, this implies that $\Pr[\mathsf{Enc}_k(m_1) = c] = 0$. By the same logic, $\Pr[\mathsf{Enc}_k(m_2) = c] = 0$. Thus, Equation (1.1) is satisfied.

Case 2: Suppose $\Pr[\mathsf{Enc}_k(M) = c] > 0$. By perfect secrecy:

$$\Pr[M = m_1] = \Pr[M = m_1 \mid \mathsf{Enc}_k(M) = c].$$

Applying Bayes' Rule to the right hand side:

$$\Pr[M = m_1] = \frac{\Pr[M = m_1] \cdot \Pr[\mathsf{Enc}_k(M) = c \mid M = m_1]}{\Pr[\mathsf{Enc}_k(M) = c]}.$$

We can cancel out the $\Pr[M = m_1]$ terms because $\Pr[M = m_1] > 0$. Simplifying $\Pr[\mathsf{Enc}_k(M) = c \mid M = m_1]$ as above, we get that

$$1 = \frac{\Pr[\mathsf{Enc}_k(m_1) = c]}{\Pr[\mathsf{Enc}_k(M) = c]}.$$

Applying the same argument to m_2 implies:

$$1 = \frac{\Pr[\mathsf{Enc}_k(m_2) = c]}{\Pr[\mathsf{Enc}_k(M) = c]}.$$

Therefore $\Pr[\mathsf{Enc}_k(m_1) = c] = \Pr[\mathsf{Enc}_k(M) = c] = \Pr[\mathsf{Enc}_k(m_2) = c]$, satisfying Equation (1.1) as required.

(PI $\Rightarrow$ PS) Suppose Π satisfies perfect indistinguishability. Let M be a random variable over $\mathcal{M}$, $m \in \mathcal{M}$, and $c \in \mathcal{C}$. If $\Pr[\mathsf{Enc}_k(M) = c] = 0$ or $\Pr[M = m] = 0$ the perfect secrecy condition is satisfied.

Suppose $\Pr[\mathsf{Enc}_k(M) = c] > 0$ and $\Pr[M = m] > 0$. We must show that

$$\Pr[M = m \mid \mathsf{Enc}_k(M) = c] = \Pr[M = m].$$

Applying Bayes' Rule to the left side and simplifying as above, we have

$$\begin{aligned}\Pr[M = m \mid \mathsf{Enc}_k(M) = c] &= \frac{\Pr[M = m] \cdot \Pr[\mathsf{Enc}_k(M) = c \mid M = m]}{\Pr[\mathsf{Enc}_k(M) = c]} \\ &= \Pr[M = m] \cdot \frac{\Pr[\mathsf{Enc}_k(m) = c]}{\Pr[\mathsf{Enc}_k(M) = c]}. \qquad (1.2)\end{aligned}$$

It suffices to show that $\Pr[\mathsf{Enc}_k(M) = c] = \Pr[\mathsf{Enc}_k(m) = c]$. Applying Law of Total Probability, we get:

$$\begin{aligned}\Pr[\mathsf{Enc}_k(M) = c] &= \sum_{m' : \Pr[M = m'] > 0} \Pr[M = m'] \cdot \Pr[\mathsf{Enc}_k(M) = c \mid M = m'] \\ &= \sum_{m' : \Pr[M = m'] > 0} \Pr[M = m'] \cdot \Pr[\mathsf{Enc}_k(m') = c].\end{aligned}$$

By perfect indistinguishability, $\Pr[\mathsf{Enc}_k(m') = c] = \Pr[\mathsf{Enc}_k(m) = c]$ for all $m' \in \mathcal{M}$. Continuing from the previous step, we get

$$\Pr[\mathsf{Enc}_k(M) = c] = \Pr[\mathsf{Enc}_k(m) = c] \cdot \sum_{m' : \Pr[M=m'] > 0} \Pr[M = m'] = \Pr[\mathsf{Enc}_k(m) = c].$$

Plugging this into the denominator on the RHS of Equation (1.2) completes the proof:

$$\Pr[M = m \mid \mathsf{Enc}_k(M) = c] = \Pr[M = m]. \qquad \square$$

1.4 One-time pad (OTP): a perfectly secret scheme

For any message length $\ell \geq 1$, the one-time pad (OTP) for length ℓ messages OTP_ℓ has message space, ciphertext space, and key space $\mathcal{M} = \mathcal{C} = \mathcal{K} = \{0,1\}^\ell$. Encryption and decryption are as follows.

$\mathrm{OTP}_\ell.\mathsf{Enc}_k(m)$	$\mathrm{OTP}_\ell.\mathsf{Dec}_k(c)$
return $k \oplus m$	**return** $k \oplus c$

Claim 1.4. *For every message length ℓ, OTP_ℓ is a correct symmetric encryption scheme.*

Proof. Exercise 1.2. $\square$

Theorem 1.2
For any message length ℓ, OTP_ℓ is perfectly secret.

Proof. By Theorem 1.1, it suffices to prove perfect indistinguishability. Let $m, c \in \{0,1\}^\ell$ be arbitrary.

$$\Pr_{k \xleftarrow{\$} \{0,1\}^\ell}[\mathsf{Enc}_k(m) = c] = \Pr_k[k \oplus m = c] = \Pr_k[k = c \oplus m] = 2^{-\ell}.$$

Hence, perfect indistinguishability is satisfied: for all $m_1, m_2 \in \{0,1\}^\ell$ and $c \in \{0,1\}^\ell$,

$$\Pr[\mathsf{Enc}_k(m_1) = c] = 2^{-\ell} = \Pr[\mathsf{Enc}_k(m_2) = c].$$

$\square$

1.4.1 Limitations of OTP

Long keys Notice that OTP keys are the same length as messages. For Alice to send an ℓ-bit secret message m to Bob, she must first send an ℓ-bit secret key k to Bob. This is called **key distribution**, and is the major obstacle to using OTP in practice.

Still, there are situations where the OTP can be useful. For example, Alice might distribute a secret key by some slow and expensive means. For example, Alice (or the USSR) might send a code book with a 100,000 bit long key by courier. That key enables real-time secure communication of 100,000 plaintext bits, piece by piece over time. For the most sensitive communications, the expense of OTP key distribution may be warranted.

Keys can't be reused Long keys wouldn't be a problem if they could be reused many times. But they can't. Perfect secrecy implies that OTP is immune to attack as long as the key is only used once. If keys are reused, all bets are off. See Exercise 1.8. The best illustration of this that I know of is visual: `https://crypto.stackexchange.com/a/108`. For an interactive demo using text, and how this led to decrypting some Soviet communications, see `https://ciphermuseum.com/ciphers/venona.html`.

★ **Ciphertexts are "malleable"** Perfect secrecy protects confidentiality, but does not guarantee that a ciphertext has not been modified. If Eve is able to modify the ciphertext in transit, then she can change the message that Bob receives. This is true even if Eve doesn't know the message itself!

For example, Eve might know that Alice is trading NVIDIA stock. Her orders start with the string `NVDA`, followed by 100 additional characters. Suppose characters are encoded using 8-bit ASCII. Write the message as $m = u \| r$, where u is the 32-bit encoding of `NVDA`, $r \in \{0,1\}^{800}$ is unknown to Eve, and $\|$ denotes concatenation. Eve wants to change the stock to `AAPL` while leaving the rest of the order intact. Let v be the 32-bit encoding of `AAPL`. Eve computes a mask $u \oplus v$:

$u =$	01001110	01010110	01000100	01000001
$v =$	01000001	01000001	01010000	01001100
$u \oplus v =$	00001111	00010111	00010100	00001101

Upon intercepting c, Eve XORs its first 32 bits with this mask and leaves the remaining 800 bits unchanged:

$$\hat{c} = c \oplus \big((u \oplus v) \parallel 0^{800}\big).$$

Bob decrypts this ciphertext to

$$\mathsf{Dec}_k(\hat{c}) = k \oplus \hat{c} = m \oplus \big((u \oplus v) \parallel 0^{800}\big) = u \oplus (u \oplus v) \parallel r \oplus 0^{800} = v \parallel r.$$

Eve has changed the stock without learning the key or the remaining 800 bits. More generally, replacing c by $c \oplus \Delta$ changes the decrypted message to $m \oplus \Delta$.

1.5 Long keys are necessary for perfect secrecy

The OTP is perfectly secret, but suffers from a major drawback: keys are as long as messages. Unfortunately, this is inherent. Any perfectly secret encryption scheme must have as many keys as messages $|\mathcal{K}| \geq |\mathcal{M}|$. If keys and messages are bit strings of fixed length, this implies that $|k| \geq |m|$.

Theorem 1.3 (Shannon's Theorem)
Let Π be a symmetric encryption scheme with message space $\mathcal{M}$ and key space $\mathcal{K}$. If Π is perfectly secret and correct, then $|\mathcal{K}| \geq |\mathcal{M}|$.

Proof. Let $m^* \in \mathcal{M}$ be any message. Let $c^* \in \mathcal{C}$ be an arbitrary ciphertext that could possibly result from encrypting m^*. Namely, $\Pr[\mathsf{Enc}_k(m^*) = c^*] > 0$. Let $\mathcal{M}_{c^*}$ be all messages that could possibly result from decrypting c^*, using every possible key:

$$\mathcal{M}_{c^*} := \{m \in \mathcal{M} \ : \ \exists k \in \mathcal{K},\ m = \mathsf{Dec}_k(c^*)\}.$$

By correctness of Π, for each $k \in \mathcal{K}$, there is at most 1 message m' such that $\mathsf{Dec}_k(c^*) = m'$. Hence

$$|\mathcal{M}_{c^*}| \leq |\mathcal{K}|.$$

By perfect secrecy, for every message $m \in \mathcal{M}$,

$$\Pr[\mathsf{Enc}_k(m) = c^*] = \Pr[\mathsf{Enc}_k(m^*) = c^*] > 0.$$

Thus, there exists a key k_m such that $\Pr[\mathsf{Enc}_{k_m}(m) = c^*] > 0$. By correctness, $\mathsf{Dec}_{k_m}(c^*) = m$. Therefore $m \in \mathcal{M}_{c^*}$ and $\mathcal{M} = \mathcal{M}_{c^*}$. Putting it all together completes the proof:

$$|\mathcal{M}| = |\mathcal{M}_{c^*}| \leq |\mathcal{K}|. \qquad \square$$

1.6 Exercises

Exercise 1.1. What is $|\mathsf{SUB}.\mathcal{K}|$? For key length $\ell \geq 2$, what is $|\mathsf{VIG}.\mathcal{K}|$?

Exercise 1.2. Prove that SUB, VIG, and OTP_ℓ are correct encryption schemes (for any $\ell > 0$).

Exercise 1.3 (A correct but insecure scheme)**.** Let $\mathcal{M} = \{0,1\}^\ell$, $\mathcal{K} = \{0,1\}^\ell$, and $\mathcal{C} = \{0,1\}^\ell \times \{0,1\}^\ell$. Consider the following encryption algorithm $\mathsf{Enc}_k(m) = (k,\ k \oplus m)$.

(a) Describe in pseudocode a decryption algorithm Dec such that the scheme $(\mathsf{Enc}, \mathsf{Dec})$ is correct.

(b) Show how to compute m given $\mathsf{Enc}_k(m)$. Therefore, this scheme is not secure.

Exercise 1.4. Definition 1.2 requires Dec to be deterministic. Show that this choice is without loss of generality.

Hint: Suppose that $(\mathsf{Enc}, \mathsf{Dec})$ *is an encryption scheme with randomized* Dec*. Construct an equivalent encryption scheme* $(\mathsf{Enc}', \mathsf{Dec}')$ *with deterministic* Dec'*. Use correctness (Def. 1.2).*

Exercise 1.5. Assume an attacker knows that Alice encrypted one of two messages: $m_0 =$ `ABCD` or $m_1 =$ `AKTD`. Suppose the attacker learns that c is an encryption of the message using the Vigenère cipher with key length $\ell \geq 2$. For what values of ℓ can the attacker determine the message with certainty? Justify your answer.

Exercise 1.6. Let M and Enc be defined as in Example 1.5. Let $c =$ `NCO QPEZ`. Compute the following:

(a) $\Pr[\mathsf{Enc}_k(M) = c]$

(b) $\Pr[M =$ `SEL NVDA` $\mid \mathsf{Enc}_k(M) = c]$

(c) $\Pr[M = \texttt{BUY NVDA} \mid \mathsf{Enc}_k(M) = c]$

Exercise 1.7. Prove that OTP is perfectly secret (Theorem 1.2).

Exercise 1.8. The following questions concern multiple encryptions of single-character ASCII plaintexts with the one-time pad using the same 8-bit key. You may assume that the plaintexts are either (upper- or lower-case) English letters or the space character.

(a) Say you see the ciphertexts $C_1 = 1011\ 0111$ and $C_2 = 1110\ 0111$. What can you deduce about the plaintext characters these correspond to?

(b) Say you see the ciphertexts $C_1 = 0110\ 0110$, $C_2 = 0011\ 0010$, and $C_3 = 0010\ 0011$. What can you deduce about the plaintext characters these correspond to?

Hint: Focus on the second bit of the ciphertexts.

Exercise 1.9. Describe in pseudocode an encryption scheme Π satisfying all of the following. You may choose $\mathcal{M}$, $\mathcal{K}$, $\mathcal{C}$ however you'd like, as long as they're finite and nonempty.

- Π is perfectly secret.
- Every ciphertext is possible: For every $c \in \mathcal{C}$, there exists $m \in \mathcal{M}$, $k \in \mathcal{K}$ such that $\Pr[\mathsf{Enc}_k(m) = c] > 0$.
- Π doesn't have uniform ciphertexts: For every $m \in \mathcal{M}$, and uniformly sampled key $k \xleftarrow{\$} \mathcal{K}$ the ciphertext $\mathsf{Enc}_k(m)$ is not uniformly distributed over $\mathcal{C}$.

Hint: OTP satisfies the first two properties. Modify OTP to make Enc *randomized and produce non-uniform ciphertexts.*

Exercise 1.10. One possible key for the ℓ-bit one-time pad is the all-zero string 0^ℓ. When this key is used, one-time pad encryption simply outputs the message m without changing it. A natural suggestion is to modify the scheme so that the key is never set to 0^ℓ.

Fix $\ell > 0$ arbitrary. Let $\mathcal{M} = \mathcal{K} = \mathcal{C} = \{0,1\}^\ell$ and let OTP_ℓ be the ℓ-bit one-time pad encryption scheme.

(a) Let $\mathcal{M}' = \mathcal{M}$, $\mathcal{C}' = \mathcal{C}$ and $\mathcal{K}' = \mathcal{K} \setminus \{0^\ell\}$. Define the encryption scheme $\mathrm{OTP}'_\ell = \mathrm{OTP}_\ell$ for $\mathcal{M}', \mathcal{C}', \mathcal{K}'$.

Prove that OTP'_ℓ is not perfectly secret.

Hint: There is a one line solution using Theorem 1.3.

(b) This part is identical to the previous, except we also remove the message 0^ℓ from the message space.

Let $\mathcal{M}'' = \mathcal{M} \setminus \{0^\ell\}$, $\mathcal{C}'' = \mathcal{C}$ and $\mathcal{K}'' = \mathcal{K} \setminus \{0^\ell\}$. Define the encryption scheme $\mathrm{OTP}''_\ell = \mathrm{OTP}_\ell$ for $\mathcal{M}'', \mathcal{C}'', \mathcal{K}''$.

Prove that OTP''_ℓ is not perfectly secret.

Exercise 1.11 (★ Two-message secrecy, non-malleability (Updated 10/6/26)**.** OTP has the following property: For any m and c, there is a unique key k such that $\mathsf{Enc}_k(m) = c$. This problem considers an encryption scheme satisfying a version of this that holds for any two message-ciphertext pairs.

Let p be a prime. Consider an encryption scheme with message-, ciphertext-, and key-spaces

$$\mathcal{M} = \mathcal{C} = \{m : 0 \le m < p\},$$
$$\mathcal{K} = \{(a, b) : 1 \le a < p,\ 0 \le b < p\}.$$

The encryption algorithm is $\mathsf{Enc}_k(m) = [am + b \bmod p]$, where $k = (a, b)$.

(a) Compute $|\mathcal{M}|$ and $|\mathcal{K}|$.

(b) (Added 10/6/26) Describe a decryption algorithm so that the resulting encryption scheme is perfectly correct.

(c) Prove that for every pair of distinct messages $m_1 \neq m_2$ and distinct ciphertexts $c_1 \neq c_2$, there exists a unique key k such that: $\mathsf{Enc}_k(m_1) = c_1$ and $\mathsf{Enc}_k(m_2) = c_2$.

Hint: Solve the system of equations $am_1 + b = c_1 \pmod p$ *and* $am_2 + b = c_2 \pmod p$*. Use the fact that every* $x \neq 0 \pmod p$ *has a multiplicative inverse* x^{-1} *modulo* p *(i.e.,* $x \cdot x^{-1} = 1 \pmod p$*).*

(d) Prove the scheme is perfectly secret.

Hint: Prove perfect indistinguishability.

(e) The OTP was malleable in the following sense: There exists a non-identity function F (i.e., $F(x) = x \oplus \Delta$) such that

$$\mathsf{Dec}_k(F(\mathsf{Enc}_k(m))) = F(m).$$

Explain why the property proved in part (2) of this problem implies that, for the present scheme, no such function exists.

Exercise 1.12 (★ Adapted from [Katz and Lindell, 2021], Exercise 2.19)**.** One could try to define a two-message version of perfect secrecy as follows.

> **Definition 1.5** (Two-message perfect secrecy (impossible!))**.** Π *is* perfectly secret for two messages *if for all jointly-distributed random variables* (M_1, M_2) *over pairs of messages* $\mathcal{M} \times \mathcal{M}$*, all messages* $m_1, m_2 \in \mathcal{M}$*, and all ciphertexts* $c_1, c_2 \in \mathcal{C}$*:*
>
> $$\Pr\Big[M_1 = m_1 \wedge M_2 = m_2 \;\Big|\; \mathsf{Enc}_k(M_1) = c_1 \wedge \mathsf{Enc}_k(M_2) = m_2\Big] = \Pr[M_1 = m_1 \wedge M_2 = m_2],$$
>
> *or*
>
> $$\Pr[\mathsf{Enc}_k(M_1) = c_1 \wedge \mathsf{Enc}_k(M_2) = c_2] = 0.$$

Prove that the above definition is impossible to satisfy.

Hint: Take $c_1 = c_2$*.*

Exercise 1.13 (Adapted from Exercise 2.8 from [Katz and Lindell, 2021])**.** State whether the following encryption scheme is perfectly secret. Justify your answer.

The message space is $\mathcal{M} = \{m \in \{0,1\}^\ell \mid \text{the last bit of m is } 0\}$. The key space is $\mathcal{K} = \{0,1\}^{\ell-1}$. $\mathsf{Enc}_k(m)$ returns the ciphertext $m \oplus (k\|0)$, and $\mathsf{Dec}_k(c)$ returns $c \oplus (k\|0)$.

Exercise 1.14 (Exercise 2.9 from [Katz and Lindell, 2021])**.** In each of the following schemes, $\mathsf{Enc}_k(m) = [m+k \bmod 3]$. State in each case whether the scheme is perfectly secret, and justify your answers.

(a) $\mathcal{M} = \{0,1\}$ and $\mathcal{K} = \{0,1\}$.

(b) $\mathcal{M} = \{0,1,2\}$ and $\mathcal{K} = \{0,1,2\}$.

(c) $\mathcal{M} = \{0,1\}$ and $\mathcal{K} = \{0,1,2\}$.

(d) $\mathcal{M} = \{0,1,2\}$ and $\mathcal{K} = \{0,1\}$ *Hint: Use Shannon's Theorem.*

Exercise 1.15 (Exercise 2.10 from [Katz and Lindell, 2021])**.** The following questions concern the message space $\mathcal{M} = \{0,1\}^{\leq \ell}$, the set of all nonempty binary strings of length at most ℓ.

(a) Consider the encryption scheme in which $\mathcal{K} = \{0,1\}^\ell$ and $\mathsf{Enc}_k(m)$ outputs $k_{1:|m|} \oplus m$, where $k_{1:t}$ denotes the first t bits of k. Show that this scheme is not perfectly secret for message space $\mathcal{M}$.

(b) (★) Design a perfectly secret encryption scheme for $\mathcal{M}$.

Bibliography

[Katz and Lindell, 2021] Katz, J. and Lindell, Y. (2021). *Introduction to Modern Cryptography*. Chapman and Hall/CRC, 3rd edition.